Containing Epidemics on the Global Air-Transport Network: Budget, Centrality, and the Epidemic Threshold

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Abstract—We study how limited airport closures affect epidemic spreading on the global air-transport network. On a cleaned OpenFlights graph of 3,188 airports, we compare random closure against degree-, closeness-, betweenness-, and eigenvector-targeted strategies under equal budgets of 1–20%, simulated with an Euler-integrated SIR metapopulation model. At a 20% budget, targeted strategies cut the final outbreak to 1.8–10.1%, versus 61.6% for random closure, but fragment the network: the strategy that best suppresses the epidemic is not the one that best preserves connectivity.

I. Motivation / Problem Statement

Air travel connects distant populations and can accelerate geographic epidemic spread. When an outbreak starts, closing every airport is not realistic – it would stop trade, essential travel, and emergency movement. The practical question is therefore not whether to close airports, but which airports to close given a limited budget. This work answers that question for the global air-transport network by comparing different closure strategies under the same budget, initial infection, and model assumptions. [1], [2]

II. Methodology

A. Network construction

We use the OpenFlights airport and route files downloaded on 27 October 2026 [3]. Airports are nodes and routes are undirected edges: if a route exists between two airports, we assume people can travel in both directions. Invalid numeric identifiers, self-loops (an airport connected to itself), duplicate routes, and isolated airports with no connections are removed. The cleaned graph has 3,214 airports and 18,858 connections in seven components; its largest component has 3,188 airports and 18,833 connections. We run the epidemic experiments on this largest component because an epidemic can only spread inside a connected part of the network. Table I summarizes the graph used for the epidemic experiments.

TABLE I
OpenFlights network summary

Measure	Value
Airports (full graph)	3,214
Routes (full graph)	18,858
Largest-component airports	3,188
Largest-component routes	18,833
Average degree (largest component)	11.81
Connected components	7

B. Centrality and interventions

We calculate degree, closeness, betweenness, and eigenvector centrality using NetworkX, NumPy, pandas, and Matplotlib [4], [5], [6], [7]. Degree counts direct connections; closeness measures average distance to all other airports; betweenness counts shortest paths passing through an airport; eigenvector rewards connections to other important airports. For a budget b, the top $\lceil b|V| \rceil$ airports are removed for a centrality strategy. Random closure samples the same number of airports using a fixed seed. Table II lists the top airports identified by each centrality measure.

TABLE II
Top airports by centrality in the largest component

Measure	Top three airports (OpenFlights ID)
Degree	Schiphol (580), Frankfurt (340), Charles de Gaulle (1382)
Closeness	Frankfurt (340), Charles de Gaulle (1382), Heathrow (507)
Betweenness	Charles de Gaulle (1382), LAX (3484), Anchorage (3774)
Eigenvector	Schiphol (580), Frankfurt (340), Charles de Gaulle (1382)

C. Related work

Network epidemic models study how diseases spread over contact graphs rather than well-mixed populations. Keeling and Eames survey how network structure affects epidemic thresholds and final sizes [1], while Pastor-Satorras et al. review epidemic processes on complex networks, showing that heterogeneous structure changes how diseases spread [2]. This project applies these ideas to the real global air-transport network: airports are contact points, routes are possible transmission paths, and we test whether targeting structurally important airports controls an outbreak better than random closure.

D. SIR threshold

The normalized mean-field model is

$$\dot{S} = -\beta SI, \quad \dot{I} = \beta SI - \gamma I, \quad \dot{R} = \gamma I. \tag{1}$$

Here S, I, and R are the fractions of susceptible, infected, and recovered people. β is how fast infection spreads per contact, and γ is how fast infected people recover. The

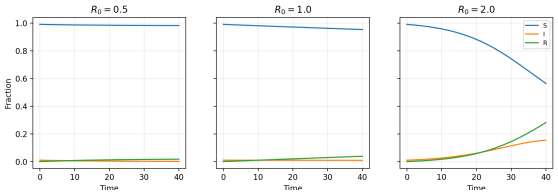


Fig. 1. Mean-field SIR behavior below, at, and above the epidemic threshold.

key quantity is $R_0 = \beta S_0/\gamma$: if it is below 1, each infected person infects fewer than one person on average and the outbreak dies out; if it is above 1, the outbreak can grow. With $\beta = 0.30$ and $\gamma = 0.10$, we get $R_0 = 3 > 1$, so under our assumptions an outbreak can spread. Figure 1 shows the mean-field curves below, at, and above this threshold.

E. Network metapopulation model

The simple SIR model describes one population. Our network model runs an SIR system at every airport and lets people move between connected airports. Each airport has susceptible, infected, and recovered populations. Infection spreads locally at rate $\beta S_i I_i / N_i$, people recover at rate γI_i, and an undirected graph-diffusion term couples neighbouring airports. For each compartment $X_i \in \{S_i, I_i, R_i\}$, mobility adds $\mu\left(\sum_{j \in N(i)} X_j - k_i X_i\right)$, where k_i is airport i's degree: people leave airport i and arrive at its neighbours, carrying their infection status with them. We integrate with the explicit Euler method using $\beta = 0.30$, $\gamma = 0.10$, mobility rate 0.02, normalized population one per airport, $dt = 0.1$, and duration 80. The initial infected fraction is 1% at one fixed high-degree seed airport, which is protected from closure so every strategy starts from the same condition. For immediate closure, selected airports and their routes are removed before simulation. For delayed closure, mobility edges to closed airports are removed and local infection there is set to zero at the intervention time; existing populations remain and can recover.

F. Interventions and measurements

We compare random, degree, closeness, betweenness, and eigenvector closures for budgets from 1% to 20%. Outcomes include final outbreak size, peak infection, time to peak, and largest connected component. Final outbreak size is the cumulative recovered fraction at the end of the simulation. We define containment as final outbreak size at most 5%. The complete sweep evaluates every integer budget from 1% to 20%; random closure uses five independent trials because a single random draw could be unusually lucky or unlucky.

III. Results and Key Findings

Figure 2 shows the complete generated real-network comparison. At a 20% budget, closeness has the smallest final outbreak size (1.80%), followed by degree (3.00%),

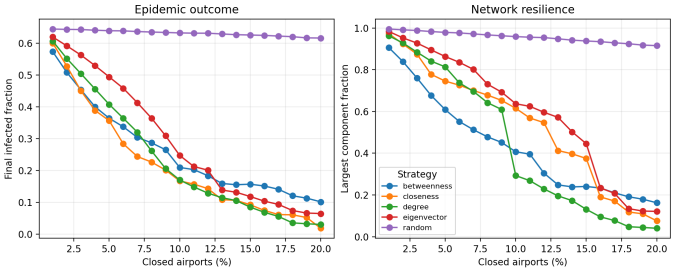


Fig. 2. Complete 1–20% closure-budget results on the largest OpenFlights component.

TABLE III
Final outbreak and connectivity at a 20% budget

Strategy	Final outbreak	Peak infected	LCC fraction
Random	61.63%	25.45%	91.55%
Degree	3.00%	0.80%	4.04%
Closeness	1.80%	0.59%	7.57%
Betweenness	10.11%	3.22%	16.24%
Eigenvector	6.43%	2.22%	12.12%

eigenvector (6.43%), betweenness (10.11%), and random closure (61.63%). Degree reaches the 5% containment criterion first, at 18

What these numbers mean: random closure removes airports at random, so even at a 20% budget it misses many of the airports that matter most for transmission, leaving 61.63% of the population eventually infected. Targeted strategies remove structurally important airports – hubs with many connections, or airports that lie on many shortest paths – so they block far more transmission paths per closed airport. Closeness targets airports that can reach the rest of the network quickly, which is why it gives the smallest final outbreak at 20%. The trade-off is connectivity: closing the most important airports breaks the network apart. Degree closure leaves only 4.04% of remaining airports in the largest connected component at 20%, meaning most of the surviving airports are in small, disconnected pieces.

The compact synthetic comparison was used as a short-horizon implementation check across ER, BA, and WS networks at a 10% closure budget. Because these validation runs ended at time 20 while infection was still active, they are not used for the main containment conclusion. Figure 3 shows the synthetic results. The degree-preserving null comparison retained almost the same random-closure connectivity as the real network (96.5% versus 96.6

Delayed intervention was evaluated with a fixed 10% betweenness closure budget. With immediate, 20-unit, and 40-unit intervention, final outbreak sizes were 18.83%, 54.15%, and 57.02%, respectively, and peak infected fractions were 6.30%, 23.21%, and 24.44%. Thus, intervening later substantially increases outbreak size in this model: once the infection has spread widely, closing airports later

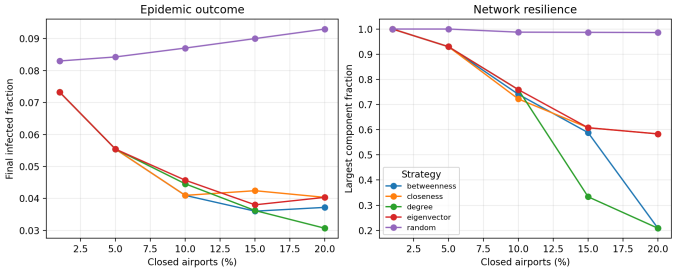


Fig. 3. Synthetic closure-budget comparison across ER, BA, and WS networks at a 10% budget.

cannot undo the early spread.

IV. Discussion / Analysis

The real network is heterogeneous: a small set of airports has a large role in connecting otherwise weakly connected regions. Degree and closeness performed best for reducing the cumulative outbreak under the tested parameters, whereas betweenness caused strong fragmentation but did not achieve the 5% containment criterion by 20%. Anchorage ranks third on betweenness because many intercontinental shortest paths pass through high-latitude connections, yet it is not the most valuable target for suppressing transmission in this model: structural importance for connectivity and epidemic suppression need not be identical. The appropriate strategy depends on whether the objective prioritizes epidemic reduction, connectivity preservation, or a weighted combination of both. A real health authority closing airports would need to balance lives saved against the economic and social cost of disconnecting cities.

The model has several limitations. OpenFlights route records do not provide passenger populations or time-varying traffic, so normalized airport populations and unweighted routes are used. The SIR parameters are scenario parameters rather than estimates for a particular disease. Finally, the intervention comparison measures structural removal and does not model rerouting or behavioral adaptation – for example, if an airport closes, travelers may find alternative routes, which the model does not capture.

V. Conclusion

This project provides a reproducible framework[1] for comparing airport closure strategies on a real flight network. The mean-field model validates the epidemic threshold ($R_0 = 3 > 1$ under our parameters), while the metapopulation model connects local SIR dynamics to global network structure. Degree reached the 5% containment criterion first, at 18% closure, while closeness produced the smallest final outbreak at 20% (1.80%); both achieved this by strongly fragmenting the network, so effective control must be balanced against connectivity loss.

[1]Code and generated results: https://github.com/krish-wp/PON-Project.

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